

States of Matter

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INTRODUCTION

Gas is the state of matter in which molecules are always in random motion. Intermolecular interactions are extremely small (almost negligible) as compared to other states like solid and liquid.

Gases are highly compressible state of matter and its rate of diffusion is maximum. Gases have their own importance in living world. Air is a gaseous mixture of oxygen, nitrogen, CO_2 , Ar, etc.

INTERMOLECULAR FORCES

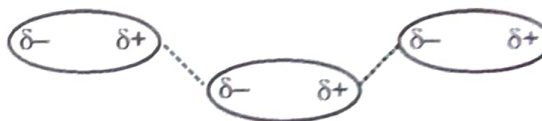
Other Forces of Attraction Present in Molecules

van der Waals Forces

These are the weak forces of attraction between two molecules with or without any strong bond. These are electrostatic in nature.

Types of van der Waals forces

1. Dipole-dipole interaction
 2. Dipole-induced dipole interaction
 3. Instantaneous dipole interaction
1. **Dipole-dipole interaction** : These forces exist between two molecules which are polar in nature. Opposite charges of two dipoles attract each other and produce interactions called **keesom forces**. In HCl, dipole-dipole interactions exist.



- (a) **Stationary polar molecules** : Dipole-dipole interaction energy is inversely proportional to the third power of the distance between stationary polar molecules (in solids)

$$\text{Interaction energy} \propto \frac{1}{r^3}$$

Dipole-dipole interaction energy is inversely proportional to the sixth power of the distance between the rotating polar molecules (like water)

$$\text{Interaction energy} \propto \frac{1}{r^6}$$

2. **Dipole-induced dipole interaction** : When a polar molecule comes closer to a non-polar molecule it induces weak polarity (dipole) in that molecule. Now, weak interactions develop between

polar molecule and molecule in which polarity is induced. These interactions are known as Debye interactions.



$$\text{Interaction energy} \propto \frac{1}{r^6}$$

3. **Instantaneous dipole interaction or London dispersion forces** : Distribution of electrons is not symmetrical (uniform) at all times in a molecule. At an instant, more electrons are gathered on one side and produce polarity in that molecule. This polar molecule produces polarity in other molecule. Now, weak interaction arises between these instantaneous dipoles. These interactions are known as Dispersion forces or London forces.



$$\text{Interaction energy} \propto \frac{1}{r^6}, \text{ where } r \text{ is the}$$

internuclear distance between two interacting particles.

These forces exist between noble gases, S_8 molecules, iodine molecules, graphite layer.

Liquefaction of gases proved the presence of van der Waals forces.

Illustration 1

The correct order of strength of intermolecular forces is correctly given in

- (1) H-bonding > London forces > Dipole – Induced dipole forces
- (2) London forces > Dipole-induced dipole forces > H-Bonding
- (3) H-Bonding > Dipole-induced dipole forces > London forces
- (4) Dipole-induced dipole forces > H-bonding > London forces

Solution : H-Bonding is strongest and London forces is the weakest among the given forces.

Answer (3)

Illustration 2

For dipole-induced dipole interaction, interaction energy is inversely proportional to r^m and for London dispersion forces, interaction energy is inversely proportional to r^n , then the value of $\frac{m}{n}$ is equal to _____

Solution : For Dipole-Induced dipole interaction

$$\text{Interaction energy} \propto \frac{1}{r^6}$$

For London dispersion forces

$$\text{Interaction energy} \propto \frac{1}{r^6}$$

$$\therefore m = 6, n = 6 \Rightarrow \frac{m}{n} = \frac{6}{6} = 1$$

Answer (01.00)

INTERMOLECULAR FORCES VERSUS THERMAL ENERGY

Intermolecular forces and thermal energy have an opposite effect on the motion of particles. Intermolecular forces are responsible for keeping the molecules together whereas thermal energy tends to keep the molecules apart and in the state of motion.

The net effect of intermolecular forces and thermal energy decides the state of the matter.

We know that there are three classical states of matter.

1. Gas
2. Liquid
3. Solid

- (i) **Intermolecular interactions** : Strength of intermolecular interactions is highest in solids, then in liquids and least in gases.



Increasing order of intermolecular interactions.

- (ii) **Thermal energy** : Thermal energy of the gaseous state particles is maximum, then is the thermal energy of liquid particles and least thermal energy is of solid particles.



Increasing order of thermal energy.

The two factors namely, pressure and temperature are the deciding factors during the inter-conversion of a state of a substance.

Example for changing a gas to a liquid, although the pressure is increased to increase the intermolecular interactions but the inter conversion is not possible unless the thermal energy of the molecule is reduced by lowering the temperature.

For example water is a liquid at ordinary temperature. When it is heated to 100°C it changes into steam (gas). Whereas, when it is cooled below 0°C , it changes into ice (solid).



MEASURABLE PROPERTIES OF GASES

- Temperature:** Experimental studies have shown that absolute zero on Kelvin scale is equivalent to -273.15°C on the Celsius scale. Both the Celsius and the Kelvin scales have units of equal magnitude that is one degree change in celsius is equivalent to one kelvin change.

$$\text{Thus } T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

- Pressure of a gas:** According to laws of motion, pressure is defined as force applied per unit area of surface.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \quad (\text{SI unit is } \text{Nm}^{-2} = \text{pascal})$$

Atmospheric Pressure:

- The actual value of atmospheric pressure depends on the temperature, location and weather conditions.
- Atmospheric pressure is measured using an instrument called barometer.

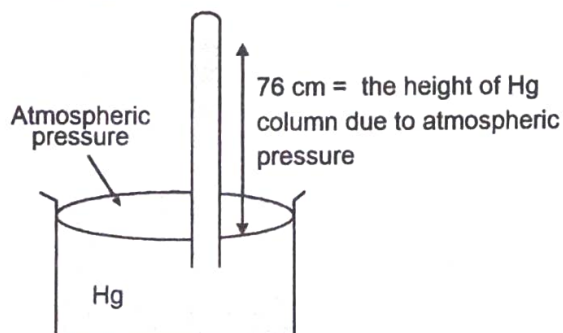


Fig.: Barometer, Measuring Atmospheric Pressure.

Standard atmospheric pressure (1.0 atm) is equal to the pressure that supports a column of mercury exactly having 76 cm height at 0°C at sea level.

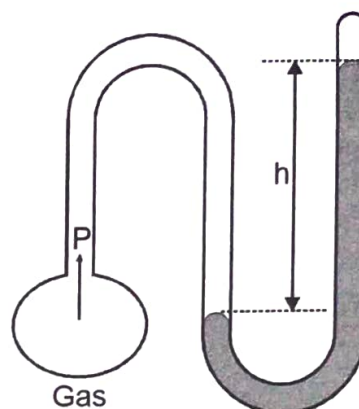
$$1 \text{ atm} = 76 \text{ cm} = 760 \text{ mm of Hg}$$

$$P = \frac{\text{Force}}{\text{Area}} = \rho gh$$

Knowledge Cloud

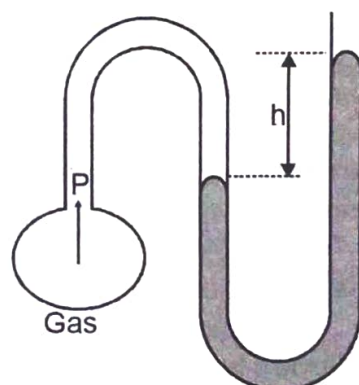
Measurement of pressure by manometer

Another device that can measure pressure is manometer. They can be either open tube manometer or closed tube manometer. In closed tube manometer, the difference in mercury levels (in mm Hg) is equal to the pressure in torr. In open tube manometer, the atmospheric pressure also has to be taken into account.



$$P_{\text{gas}} = P_h$$

Closed - end manometer



$$P_{\text{gas}} = P_h + P_{\text{atm}}$$

Open - end manometer

Instead of Hg if any other medium is used then following relation is used

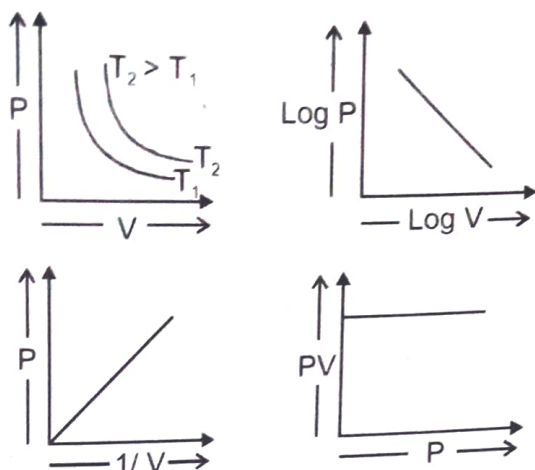
$$h_1 \rho_1 = h_2 \rho_2 (\text{Hg})$$

GASEOUS LAWS

The behaviour of gases can be explained on the basis of following laws :

All gases obey certain laws called gas laws.

Boyle's Law : Robert Boyle studied the effect of pressure on the volume of fixed mass of gases at constant temperature. He plotted several P-V isotherms for gases and got graphs like;



Interpretation: The volume of a given mass of gas is inversely proportional to pressure at constant temperature.

- $\Rightarrow V \propto \frac{1}{P}$ (Keeping T and n constant)
 $\Rightarrow PV = \text{constant}$ or $P_1V_1 = P_2V_2 = P_3V_3 \dots$ and so on.
 $\Rightarrow \log P + \log V = \text{constant}$

Note: Pressure of water vapours does not depend on volume. It depends on temperature. If temperature is constant, it remains constant.

Relation between Pressure and Density: Boyle's law proves that the gases are compressible because when a given mass of a gas is compressed, its volume decreases. So, this means that the same number of molecules occupy the smaller space or smaller volume. Further we can say that on applying pressure the density of the gas increases.

Density of gas $\propto$ Pressure applied on it at constant temperature

According to Boyle's law,

$$P = k_1 \frac{1}{V} \quad \dots(i)$$

Density is mass per unit volume

$$\text{i.e., } d = \frac{M}{V} \quad \dots(ii)$$

where d = density of gas

M = mass of the gas

V = volume occupied by the gas

Now, putting the value of V from equation (i) in equation (ii), we get

$$d = \left(\frac{M}{k_1} \right) P$$

$$d = k' P$$

$$\text{where, } k' = \frac{M}{k_1}$$

The above equation shows that at a constant temperature, density is directly proportional to the pressure of a fixed mass of the gas.

Physical Significance of Boyle's Law: As discussed before, on increasing pressure, the density of the gas increases at constant temperature. This indicates that gases are compressible. Same effect can be seen in the daily life. Air is denser at sea level and as the altitude increases air pressure decreases, which means air now becomes less denser. So, less oxygen molecules occupy the same volume. Therefore oxygen in air becomes insufficient for normal breathing, as a result **altitude sickness** occurs with symptoms like headache, uneasiness. That is why mountaineers have to carry oxygen cylinders with them in case of emergency to restore normal breathing.

Example 1: A bulb of unknown volume V contains an ideal gas at a pressure of 1 atm. This bulb was connected to another evacuated bulb of 0.5 litre volume through a stop cock. When the stop cock is opened the pressure in each bulb became 530 mm of mercury. Find the value of V . Assume that the temperature remains constant.

Solution: Initial volume of the gas = V litre
 Final volume of the gas = $V + 0.5$ litres
 Initial pressure of the gas = 1 atm = 760 mm
 Final pressure of the gas = 530 mm
 Since temperature is constant,

$$\begin{array}{ccc} P_1V_1 & = & P_2V_2 \\ \text{(Initial)} & & \text{(Final)} \end{array}$$

$$\text{Thus, } 760 \times V = 530 \times (V + 0.5)$$

$$V + 0.5 = \frac{760}{530} \times V$$

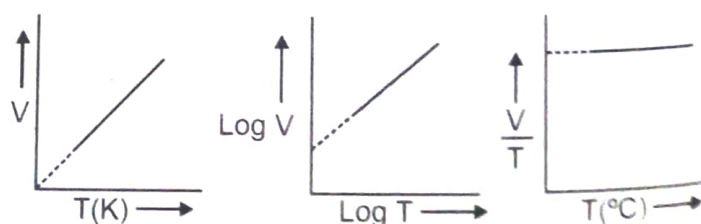
$$V + 0.5 = 1.434 V$$

$$0.434 V = 0.5$$

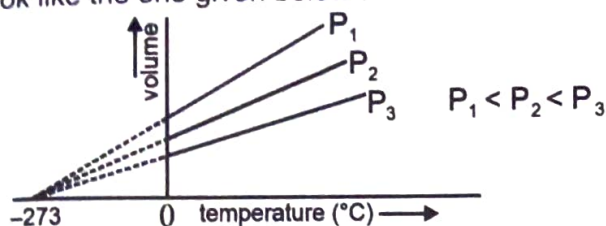
$$V = 1.152 \text{ litres}$$

Charles' Law

Jacques Charles studied the effect of temperature on volume of fixed mass of gas. He plotted several V - T isobar for gases and got the graphs as given below:



It is most important to use the Kelvin temperature here, not the Celsius scale. In Celsius scale graphs look like the one given below :



All gases expand or contract on increasing or decreasing the temperature. Experimentally it has been observed that for each degree change of temperature, the volume of sample of a gas changes by the fraction $\frac{1}{273}$ of its volume at 0°C .

Let the volume of a given amount of a gas be V_0 at 0°C . The temperature is increased by $T^\circ\text{C}$ and the new volume becomes V_t , thus

$$\Rightarrow V_t = V_0 + \frac{V_0}{273} \times T$$

$$\Rightarrow V_t = V_0 \left(1 + \frac{T}{273} \right)$$

$$\Rightarrow V_t = V_0 \left(\frac{273 + T}{273} \right)$$

$$\Rightarrow \frac{V_t}{V_0} = \frac{273 + T}{273}$$

Interpretation: The volume of a given mass of gas is directly proportional to its absolute temperature at constant pressure.

$$V \propto T \text{ (keeping } P \text{ constant)}$$

$$\Rightarrow \frac{V}{T} = \text{constant or } \frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{V_3}{T_3} = \text{constant}$$

$$\Rightarrow \log V - \log T = \text{constant}$$

The relation $\frac{V}{T} = \text{constant}$ at zero Kelvin or at -273.15°C , leads to the concept that volume of gas becomes zero. But this is not possible since a gas liquefies or solidifies much before reaching -273.15°C . The zero on kelvin scale (named after British Physicist Lord Kelvin) is known as absolute zero temperature or zero in Kelvin scale temperature or zero in thermodynamic scale of temperature.

$$T(\text{K}) = T(^{\circ}\text{C}) + 273$$

Note : Change in temperature remains same both in kelvin and $^{\circ}\text{C}$ scale.

Example 2 : Find out the percentage increase in the volume of gas when temperature of the gas is raised by 1°C , say from 0°C to 1°C at constant pressure.

Solution : Say volume of the gas at $0^\circ\text{C} = V$

According to Charles' Law, volume of the gas at 1°C

$$V_1 = V + \frac{V}{273} = \frac{274}{273} V$$

$$\text{Increase in volume} = \frac{274}{273} V - V = \frac{1}{273} V.$$

$$\% \text{ increase in volume} = \left(\frac{1}{273} \frac{V}{V} \right) \times 100 = 0.366\%$$

Alternate Method :

$$\frac{V}{T} = k$$

$$\Rightarrow \log k = \log V - \log T$$

$$\Rightarrow 0 = \frac{dV}{V} - \frac{dT}{T} \quad [\because \text{derivative is taken on both sides}]$$

$$\Rightarrow \frac{dV}{V} = \frac{dT}{T}$$

$$\Rightarrow \frac{dV}{V} = \frac{1}{273} \Rightarrow V = 0.366$$

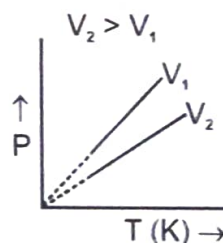
Physical Significance of Charles' Law : Hot air balloon technology is based on Charles Law. On increase in temperature air expands. So, density of air decreases. The hot air in the balloon is less dense and lighter than the atmospheric air. Therefore the balloons filled with hot air rise up for meteorological observations.

Gay Lussac's Law (Pressure-Temperature Relation)

It states that at constant volume, the pressure of a fixed mass of a gas is directly proportional to the Kelvin temperature.

The law may be expressed mathematically as

$$P \propto T \text{ or, } P = kT \text{ or, } \frac{P}{T} = \text{constant or, } \frac{P_1}{T_1} = \frac{P_2}{T_2}$$



P-T relation can be derived like V-T relation.

Physical Significance of Gay Lussac's Law :

It is seen that the pressure of the inflated tyres of automobiles is constant but in summers on a hot sunny day when the temperature is high, then the pressure inside the tyres increases, and they may burst. Similarly in winters, on a cold morning, when the temperature is low, then the pressure inside the tyres decreases considerably.

Avogadro's Law

If we add more amount of gas at constant temperature (T) and pressure (P) in a balloon containing a certain mass of gas, then the volume of gas (V) will increase

$$\therefore V \propto n.$$

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \quad \text{or} \quad \frac{V_1}{V_2} = \frac{n_1}{n_2}$$

This is the basis of Avogadro's law which states that :
"Equal volumes of all gases at same temperature and pressure contain equal numbers of molecules".

IDEAL GAS EQUATION

On combining the Boyle's law, Charles law and Avogadro's law we get an equation known as **ideal gas equation** which correlate P, V, T and moles (n) of a gas.

We have,

$$V \propto \frac{1}{P} \quad (\text{according to Boyle's law at constant } T)$$

$$V \propto T \quad (\text{according to Charles' law at constant } P)$$

$$V \propto n \quad (\text{according to Avogadro's law at constant } P \text{ and } T)$$

$$\therefore V \propto \frac{nT}{P}$$

$$PV \propto nT$$

$PV = nRT$ ideal gas equation, where R is a constant known as universal gas constant.

Numerical value of R

$$R = 0.0821 \text{ litre atm K}^{-1} \text{ mol}^{-1}$$

$$= 0.0831 \text{ litre bar K}^{-1} \text{ mol}^{-1}$$

$$= 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$= 1.987 \approx 2 \text{ cal K}^{-1} \text{ mol}^{-1}$$

$$= 8.314 \times 10^7 \text{ erg K}^{-1} \text{ mol}^{-1}$$

Other forms of ideal gas equation :

$$PV = \frac{W}{M} RT, \text{ where } W \text{ is the weight and } M \text{ is the molecular weight of the gas.}$$

$$\Rightarrow P = \frac{dRT}{M}, \text{ where } d \text{ is the density of the gas.}$$

(This is not the vapour density!)

On combining Boyle's law and Charles' law, we get

$$V = \frac{KT}{P}$$

$$\Rightarrow \frac{PV}{T} = \text{constant}$$

$$\Rightarrow \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

This is known as combined gas law.

Example 3 : 30 cm³ of a gas at 2.02 atm and 25°C was compressed to 15 cm³ at 35°C. Calculate the final pressure of the gas.

Solution : $P_1 = 2.02 \text{ atm}$

$$V_1 = 30 \text{ cm}^3$$

$$T_1 = 25^\circ\text{C}$$

$$= 25 + 273$$

$$= 298 \text{ K}$$

$$P_2 = ?$$

$$V_2 = 15 \text{ cm}^3$$

$$T_2 = 35^\circ\text{C}$$

$$= 35 + 273$$

$$= 308 \text{ K}$$

Applying, combined gas law, $\frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$

$$P_2 = \frac{(P_1 \times V_1 \times T_2)}{(V_2 \times T_1)}$$

$$\Rightarrow P_2 = \frac{(2.02 \times 30 \times 308)}{15 \times 298}$$

$$\Rightarrow P_2 = \frac{18664.8}{4470} \text{ atm}$$

$$\Rightarrow P_2 = 4.17 \text{ atm}$$

Example 4 : A balloon of diameter 20 m weighs 100 kg. Calculate its pay load when it is filled with He at 1 atm and 27°C, density of air is 1.2 kg/m³.

Solution : Weight of balloon = 100 kg = 1 × 10⁵ g

Volume of balloon

$$= \frac{4}{3} \pi r^3 = \frac{4}{3} \times \frac{22}{7} \left(\frac{20}{2} \times 100 \right)^3 = 4190 \times 10^3 \text{ L}$$

Weight of gas

$$= \frac{PVM}{RT} = \frac{1 \times 4.190 \times 10^6 \times 4}{0.0821 \times 300} = 68.13 \times 10^4 \text{ g}$$

$$\text{Total weight of gas + balloon} = 68.13 \times 10^4 + 1 \times 10^5 = 7.813 \times 10^5 \text{ g}$$

$$\text{Weight of air displaced} = \frac{1.2 \times 4190 \times 10^6}{10^3} = 5.028 \times 10^6 \text{ g}$$

$$\text{Pay load} = 5.028 \times 10^6 - 7.813 \times 10^5 = 4.246 \times 10^6 \text{ g}$$

Illustration 3

0.5 moles of gas A and x moles of gas B exert a pressure of 200 Pa in a container of volume 10 m³ at 1000 K. Given R is the gas constant in JK⁻¹ mol⁻¹, x is

$$(1) \frac{2R}{4-R}$$

$$(2) \frac{2R}{4+R}$$

$$(3) \frac{4-R}{2R}$$

$$(4) \frac{4+R}{2R}$$

Solution : $PV = nRT$ (ideal gas equation)

$$200 \times 10 \times 1000 = (0.5 + x) \times R \times 1000 \times 1000$$

$$x = \frac{4-R}{2R}$$

Answer (3)

DALTON'S LAW OF PARTIAL PRESSURE

John Dalton established the relation between pressure of the mixture of gases which do not react chemically with each other and individual pressure of the constituent gases. This relationship is known as Dalton's law of partial pressure and states that "the total pressure exerted by a mixture of non reacting gases is equal to the sum of partial pressure of each gas present in the mixture".

One gas (ideal gas) behaves like vacuum for other ideal gas.

i.e., $P_{\text{total}} = P_1 + P_2 + P_3 \dots$ (Keeping V & T constant)
where P_1 , P_2 and P_3 are partial pressure of three gases.

$$P_1 = n_1 \frac{RT}{V}$$

$$P_2 = n_2 \frac{RT}{V}$$

$$P_3 = n_3 \frac{RT}{V}$$

$$\therefore P_1 + P_2 + P_3 = (n_1 + n_2 + n_3) \frac{RT}{V} \quad \dots(i)$$

$$\text{Also, } PV = nRT$$

$$P = n \frac{RT}{V}$$

$$P = (n_1 + n_2 + n_3) \frac{RT}{V} \quad \dots(ii)$$

From (i) & (ii), we get

$$P = P_1 + P_2 + P_3$$

Dividing P_1 , P_2 , P_3 by equation (ii), we get

$$\Rightarrow \frac{P_1}{P} = \frac{n_1}{n_1 + n_2 + n_3}$$

$$\therefore P_1 = \frac{n_1}{n_1 + n_2 + n_3} \times P$$

$$\text{Similarly, } P_2 = \frac{n_2}{n_1 + n_2 + n_3} \times P$$

$$P_3 = \frac{n_3}{n_1 + n_2 + n_3} \times P$$

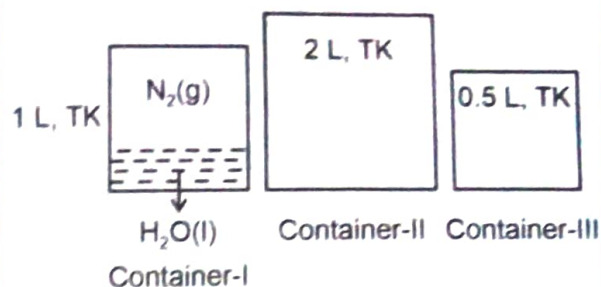
$\therefore$ Partial pressure of a gas = mole fraction $\times$ total pressure of gaseous mixture.

$\Rightarrow$ Pressure of dry gas = Pressure of moist gas – Pressure of water vapour.

= Pressure of moist gas – aqueous tension

Illustration 4

Consider the following containers :



The pressure measured over the liquid surface in the container-I is 56 mm Hg. When all the gaseous content of container-I is transferred to container-II the pressure is P_1 mm Hg and when transferred to container-III the pressure is P_2 mm Hg. The value of $\frac{P_1 + P_2}{2}$ is ____ (aqueous tension of water at TK is 16 mm Hg) (Volume of $H_2O(l)$ in container-I is negligible)

Solution : Container-II and III does not contain $H_2O(l)$ when the gaseous content is transferred to container-II,
 $P_{N_2} = \frac{56 - 16}{2} = 20$ mm Hg

As there is no $H_2O(l)$ is available to maintain the aqueous tension, the P_{H_2O} will also

reduced to $P_{H_2O} = \frac{16}{2} = 8$ mm Hg

$$\therefore P_1 = 28 \text{ mm Hg}$$

When the mixture is transferred to container-III

$$P'_{N_2} = (56 - 16) \times 2 = 80 \text{ mm Hg}$$

In this container to maintain aqueous tension $H_2O(l)$ is not required rather $H_2O(g)$ will condense to $H_2O(l)$ to maintain aqueous tension

$$P'_{H_2O} = 16$$

$$\therefore P_2 = 80 + 16 = 96 \quad \text{Answer (62.00)}$$

Example 5 : 400 mL of CO_2 gas at 0.892 bar pressure and 800 mL of CH_4 gas at 0.921 bar pressure are put into a 2L flask. Calculate the total pressure of the mixture, assuming temperature is kept constant.

Solution : Partial pressure of CO_2 in the mixture

$$P_1 V_1 = P_2 V_2$$

$$P_1 = 0.892 \text{ bar;}$$

$$P_2 = ?$$

$$V_1 = 400 \text{ mL;}$$

$$V_2 = 2 \text{ L} = 2000 \text{ mL;}$$

$$P_2 = \frac{P_1 V_1}{V_2} = \frac{0.892 \times 400}{2000} = 0.178 \text{ bar (} P_{\text{CO}_2} \text{)}$$

Partial pressure of CH_4

$$P_1 V_1 = P_2 V_2$$

$$P_1 = 0.921 \text{ bar} \quad P_2 = ?$$

$$V_1 = 800 \text{ mL} \quad V_2 = 2 \text{ L} \\ = 2000 \text{ mL}$$

$$P_2 = \frac{P_1 \times V_1}{V_2} = \frac{0.921 \times 800}{2000} = 0.368 \text{ bar (} P_{\text{CH}_4} \text{)}$$

$$\text{Total pressure of the mixture} = P_{\text{CO}_2} + P_{\text{CH}_4} \\ = 0.178 \text{ bar} + 0.368 \text{ bar}$$

$$P_{\text{total}} = 0.546 \text{ bar}$$

EXERCISE

- What will be the minimum pressure required to compress 500 cm^3 of air at 1 atm to 200 cm^3 at 30°C ?
 (1) 1 atm (2) 2 atm
 (3) 2.5 atm (4) 0.4 atm
- A gas is found to have a formula $(\text{CO})_x$. If its vapour density is 70, then value of x is
 (1) 2 (2) 3
 (3) 5 (4) 6
- 4.4 g of CO_2 contains how many litres of CO_2 at S.T.P.?
 (1) 2.4 L (2) 2.24 L
 (3) 44 L (4) 22.4 L
- At 27°C a sample of ammonia gas exerts a pressure of 5.3 atm . What is the pressure when the volume of the gas is reduced to one tenth of the original value at the same temperature?
 (1) 0.53 atm
 (2) 5.3 atm
 (3) 53 atm
 (4) None of these
- A gaseous mixture containing 2 g H_2 , 8 g He , 22 g CO_2 and 8 g O_2 is enclosed in a vessel. The gas with highest partial pressure is
 (1) He (2) H_2
 (3) CO_2 (4) O_2
- The relative strength of interionic/intermolecular forces in decreasing order is
 (1) ion-ion > ion-dipole > dipole-dipole
 (2) ion-dipole > dipole-dipole > ion-ion
 (3) ion-dipole > ion-ion > dipole-dipole
 (4) dipole-dipole > ion-dipole > ion-ion
- The predominant intermolecular forces present in ethyl acetate, a liquid, are
 (1) Dipole-dipole and hydrogen bonding
 (2) London dispersion and dipole-dipole
 (3) Hydrogen bonding and London dispersion
 (4) London dispersion, dipole-dipole and hydrogen bonding
- Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Dipole-dipole interactions are only non-covalent interactions, resulting in hydrogen bond formation.
Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.
 In the light of the above statements, choose the most appropriate answer from the options given below.
 (1) A is false but R is true
 (2) Both A and R are true and R is the correct explanation of A
 (3) A is true but R is false
 (4) Both A and R are true but R is NOT the correct explanation of A
- The pressure exerted by a non-reactive gaseous mixture of 6.4 g of methane and 8.8 g of carbon dioxide in a 10 L vessel at 27°C is _____ kPa .
 (Round off to the Nearest Integer).
 [Assume gases are ideal, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
 Atomic masses : C : 12.0 u , H : 1.0 u , O : 16.0 u]
- An LPG cylinder contains gas at a pressure of 300 kPa at 27°C . The cylinder can withstand the pressure of $1.2 \times 10^6 \text{ Pa}$. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is _____ $^\circ\text{C}$. (Nearest integer)
- A home owner uses $4.00 \times 10^3 \text{ m}^3$ of methane (CH_4) gas, (assume CH_4 is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K , mass of gas used is $x \times 10^5 \text{ g}$. The value of x is _____. (Nearest integer)
 (Given $R = 0.083 \text{ L atm K}^{-1} \text{ mol}^{-1}$)
- At 300 K , a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is _____ g mol^{-1} . (nearest integer) Assume that the behaviour of gases as ideal.
 (Given: The molar mass of hydrogen (H_2) gas is 2.0 g mol^{-1}).

13. 100 g of an ideal gas is kept in a cylinder of 416 L volume at 27°C under 1.5 bar pressure. The molar mass of the gas is _____ g mol⁻¹. (Nearest integer).
(Given : R = 0.083 L bar K⁻¹ mol⁻¹)
14. The pressure of a moist gas at 27°C is 4 atm. The volume of the container is doubled at the same temperature. The new pressure of the moist gas is _____ × 10⁻¹ atm. (Nearest integer)
(Given: The vapour pressure of water at 27°C is 0.4 atm.)
15. A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest integer)
16. 'x' g of molecular oxygen (O₂) is mixed with 200 g of neon (Ne). The total pressure of the non-reactive mixture of O₂ and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is _____.
[Given : Molar mass of O₂ = 32 g mol⁻¹.
Molar mass of Ne = 20 g mol⁻¹]

KINETIC THEORY OF GASES

The postulates of kinetic theory of gases are as follows:

- (1) The gaseous molecules are considered to be point masses.
- (2) The volume of a molecule is negligible as compared to total volume of the gas.
- (3) The molecules neither attract nor repel each other.
- (4) The collisions are perfectly elastic, i.e., there is no loss of total energy during the molecular collisions.
- (5) The average kinetic energy of molecules is directly proportional to the absolute temperature of the gas.
- (6) The effect of gravity on molecular motion is negligible.
- (7) Pressure is exerted by gas as a result of collision of particles with wall of container.
- (8) At particular time different particles have different speed and different kinetic energy. The individual speeds are changing but the distribution of speed remains constant at particular temperature.

The Kinetic Gas Equation

$$PV = \frac{1}{3} mnu^2$$

where P = Pressure of the gas

V = Volume of the gas

m = Mass of one molecule of a gas

n = number of molecules of gas

u = root mean square speed of the molecule,
for 1 mole n = N (Avogadro's number)

m × N = M = molecular weight of the gas.

For 1 mol, n = N

$$\therefore PV = \frac{1}{3} Mu^2$$

$$\Rightarrow u^2 = \frac{3PV}{M} = \frac{3RT}{M}$$

$$\Rightarrow u = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{d}}$$

Again,

$$PV = \frac{1}{3} mnu^2$$

$$PV = \frac{2}{3} \times \frac{1}{2} mNu^2 \text{ (for 1 mole)}$$

$$= \frac{2}{3} \times \frac{1}{2} mNu^2 = \frac{2}{3} \text{ K.E.}$$

$$\boxed{PV = \frac{2}{3} KT} \quad \left[\begin{array}{l} \text{since K.E.} \propto T \\ \text{K.E.} = KT \end{array} \right]$$

$$\text{Also, K.E.} = \frac{3}{2} RT \quad (\text{for 1 mole gas})$$

Note : Always pressure is calculated from the change in momentum imparted on the walls of the container.

Molecular Speeds in Gases

1. **Root mean square speed :** Maxwell proposed the term u_{rms} as the square root of mean of square of all the speeds.

$$u_{\text{rms}} = \sqrt{\frac{u_1^2 + u_2^2 + u_3^2 + \dots}{n}}$$

If there are n_1 particles with speed u_1 , n_2 particles with speed u_2 and so on, then

$$u_{\text{rms}} = \sqrt{\frac{n_1 u_1^2 + n_2 u_2^2 + n_3 u_3^2 + \dots}{n_1 + n_2 + n_3 + \dots}}$$

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3P}{d}} \text{ (for 1 mole of gas)}$$

2. **Average speed :** According to kinetic theory of gases, molecules move in different directions with different velocities.

Let n molecules be present in a given sample of gas, each one is moving with velocity $u_1, u_2, u_3, \dots, u_n$. Then,

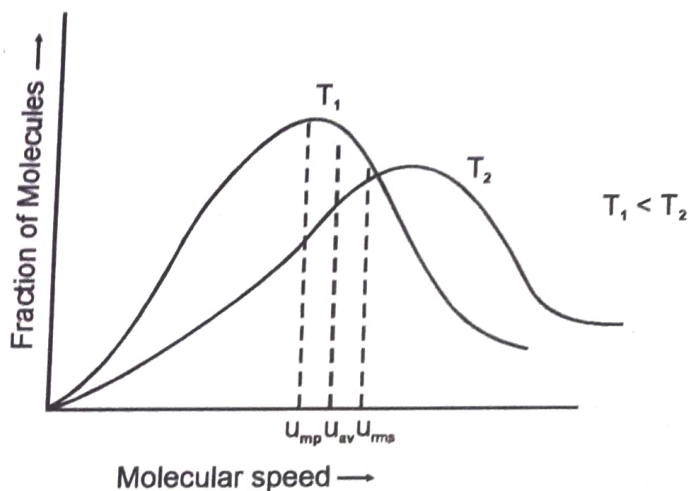
$$\text{Average speed} = \frac{u_1 + u_2 + u_3 + \dots + u_n}{n}$$

If there are n_1 particles with speed u_1 , n_2 particles with speed u_2 and so on, then

$$\text{Average speed} = \frac{n_1 u_1 + n_2 u_2 + n_3 u_3 + \dots}{n_1 + n_2 + n_3 + \dots}$$

$$\text{Mathematically, } u_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$

3. **Most probable speed** : Maxwell proposed a new term most probable speed. On the basis of probable distribution of molecular speed, a plot between fraction of molecules against their speeds give the shape as given below :



Interpretation :

- (1) A very small fraction of molecules possess either very low or very high speed.
- (2) Majority of molecules have speed in between very high and very low speeds, known as most probable speed (u_{mp}).
- (3) The numerical values of u_{mp} , u_{av} , increases with increase in temperature.
- (4) The sharpness of maxima in curves decreases with increase in temperature which reveals that number of molecules having speed in the vicinity of u_{mp} increases.
- (5) At the same temperature, lighter gases shall move faster than heavier gases as is evident

Mathematically, $u_{\text{mp}} = \sqrt{\frac{2RT}{M}}$ (This corresponds to the maxima of the curve)

from the distribution curves of chlorine and nitrogen gases in below given figure.

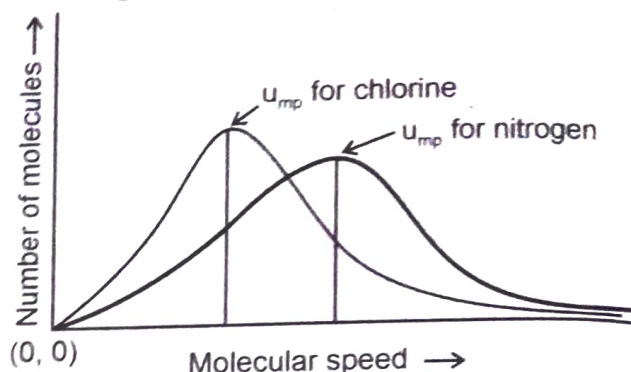


Fig. : Maxwell-Boltzmann distribution curves for chlorine and nitrogen gases at 300 K

$$(6) \quad u_{\text{mp}} : u_{\text{av}} : u_{\text{rms}} = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}} = 1 : 1.128 : 1.224$$

Thus, $u_{\text{rms}} > u_{\text{av}} > u_{\text{mp}}$

Example 6 : Calculate the pressure in atm, exerted by 10^{23} gas particles (each of mass 10^{-22} gm) in a container of volume 1 litre. The rms speed is 10^5 cm/s. What is the total kinetic energy of these particles and what must be the temperature?

$$\text{Solution : } PV = \frac{1}{3} mnC^2$$

$$P \times 10^{-3} = \frac{1}{3} \times 10^{-25} \times 10^{23} \times (10^3)^2$$

$$P = \frac{10^7}{3} \text{ N/m}^2$$

$$1.01 \times 10^5 \text{ N/m}^2 = 1 \text{ atm}$$

$$P = \frac{10^7}{1.01 \times 10^5 \times 3}$$

$$= \frac{100}{3.03} \text{ atm} = 33 \text{ atm}$$

$$\text{Total kinetic energy of molecules} = \frac{1}{2} mnC^2$$

$$= \frac{1}{2} \times 10^{-25} \times 10^{23} \times (10^3)^2$$

$$= \frac{10^4}{2} \text{ joule} = \frac{10^4}{2 \times 4.18} \text{ cal} \quad (4.18 \text{ J} = 1 \text{ cal})$$

$$= 1196.17 \text{ cal}$$

$$\text{Kinetic energy of 1 molecule} = \frac{3}{2} KT$$

$$\frac{3}{2} KT = \frac{1}{2} mC^2 \quad [n = 1]$$

$$T = \frac{mC^2}{3K}$$

$$(K = \text{Boltzmann's constant} = 1.38 \times 10^{-23})$$

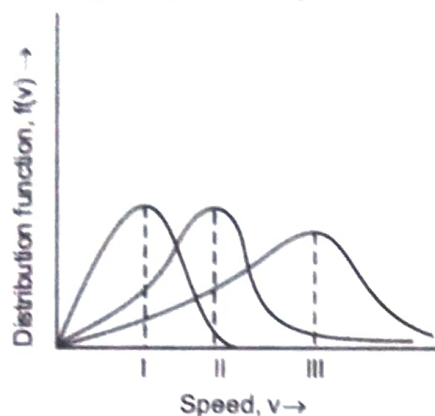
$$= \frac{10^{-25} \times (10^3)^2}{3 \times 1.38 \times 10^{-23}} = \frac{10^{-25} \times 10^6}{4.14 \times 10^{-23}}$$

$$= \frac{10^4}{4.14} = 2415 \text{ K}$$

Illustration 5

Points I, II and III in the following plot respectively correspond to

(V_{mp} : most probable velocity)



- (1) V_{mp} of N_2 (300 K); V_{mp} of O_2 (400 K); V_{mp} of H_2 (300 K)
- (2) V_{mp} of H_2 (300 K); V_{mp} of N_2 (300 K); V_{mp} of O_2 (400 K)
- (3) V_{mp} of N_2 (300 K); V_{mp} of H_2 (300 K); V_{mp} of O_2 (400 K)
- (4) V_{mp} of O_2 (400 K); V_{mp} of N_2 (300 K); V_{mp} of H_2 (300 K)

Solution : $V_{mp} = \sqrt{\frac{2RT}{M}}$

∴ As $\frac{T}{M}$ increases, V_{mp} increases

From curve,

$$(V_{mp})_I < (V_{mp})_{II} < (V_{mp})_{III}$$

$$(V_{mp})_{N_2} \propto \sqrt{\frac{300}{28}}, (V_{mp})_{O_2} \propto \sqrt{\frac{400}{32}}, (V_{mp})_{H_2} \propto \sqrt{\frac{300}{2}}$$

$$\therefore (V_{mp})_{N_2} < (V_{mp})_{O_2} < (V_{mp})_{H_2}$$

(Under given Condition) **Answer (1)**

GRAHAM'S LAW OF DIFFUSION

When two or more non-reacting gases are in contact with each other they mix spontaneously and form a homogeneous mixture. This phenomenon of mixing is known as diffusion. When a gas is allowed to escape under pressure through a fine orifice, it is called effusion.

According to Graham's Law "at constant pressure and temperature, the rate of diffusion or effusion of a gas is inversely proportional to the square root of its vapour density or molecular mass".

$$r_1 \propto \sqrt{\frac{1}{d_1}} \text{ for gas 1}$$

$$r_2 \propto \sqrt{\frac{1}{d_2}} \text{ for gas 2}$$

$$\therefore \frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}}$$

$$\Rightarrow \frac{V_1}{t_1} \times \frac{t_2}{V_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}} \text{ where, } V_1 \text{ \& } V_2 \text{ are the volume of gases diffused in time } t_1 \text{ \& } t_2.$$

$$\Rightarrow \frac{n_1}{t_1} \times \frac{t_2}{n_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}} \text{ where, } n_1 \text{ \& } n_2 \text{ are the no. of moles of gases diffused in time } t_1 \text{ \& } t_2.$$

$$\Rightarrow \frac{l_1}{t_1} \times \frac{t_2}{l_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}} \text{ where, } l_1 \text{ \& } l_2 \text{ are the distance travelled by the gases in time } t_1 \text{ \& } t_2.$$

Further rate of diffusion (r) $\propto \frac{P}{\sqrt{M}}$ at constant

temperature. Hence, $\frac{r_1}{r_2} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$ at constant temperature.

Example 7 : The volumes of ozone and chlorine diffusing during the same time are 35 ml and 29 ml respectively. If the molecular weight of chlorine is 71. Calculate molecular mass of ozone.

Solution : $\frac{r(O_3)}{r(Cl_2)} = \sqrt{\frac{M(Cl_2)}{M(O_3)}}$

Rate of diffusion in same time is proportional to volume diffused so

$$\frac{V(O_3)}{V(Cl_2)} = \sqrt{\frac{M(Cl_2)}{M(O_3)}} \Rightarrow \frac{35}{29} = \sqrt{\frac{71}{M(O_3)}}$$

$$\frac{(35)^2}{(29)^2} = \frac{71}{M(O_3)} : M(O_3) = 48.74$$

EXERCISE

17. Kinetic energy per mole of an ideal gas
 - (1) Is proportional to temperature
 - (2) Independent of temperature
 - (3) Directly proportional to the square root of temperature
 - (4) Is zero at 0°C
18. A container contains O_2 and N_2 in equal molar concentration at same temperature, what is the correct statement about the average molar kinetic energy of the two gases?
 - (1) Depends upon volume
 - (2) $KE_{N_2} = KE_{O_2}$
 - (3) $KE_{N_2} > KE_{O_2}$
 - (4) $KE_{N_2} < KE_{O_2}$

19. The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of gas, then
- (1) $T_{(H_2)} = T_{(N_2)}$ (2) $T_{(H_2)} > T_{(N_2)}$
 (3) $T_{(H_2)} < T_{(N_2)}$ (4) $T_{H_2} = \frac{T_{N_2}}{4}$
20. According to kinetic theory of gases, the gases exert the pressure due to
- (1) Collision of gas molecules
 (2) The random movement of gas molecules
 (3) The intermolecular forces of attraction between the gas molecules
 (4) The collision of gas molecules against the walls of container
21. Nitric oxide gas at 2.0 atm and oxygen gas at 1 atm are injected simultaneously through pinhole of identical area of cross section at the opposite end of a circular glass tube. The length of tube is 1.0 m and gases are kept at same temperature. Determine the distance from O_2 end at which first flash of brown fume would be observed
- (1) 0.11 m (2) 0.33 m
 (3) 0.85 m (4) 0.67 m

BEHAVIOUR OF REAL GASES

Deviation from Ideality (van der Waals Equation)

In the previous sections, we studied about the behaviour of ideal gases.

Ideal gas : A gas which obeys the gas laws and the gas equation $PV = nRT$ strictly at all temperatures and pressures is said to be an ideal gas.

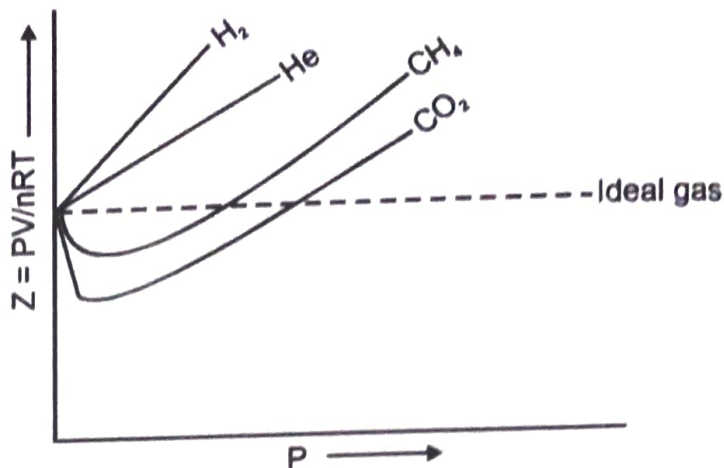
But actually the concept of ideal gas is hypothetical as there is **no gas which practically is ideal**. So, the non-ideal gases are the real gases which are the actually existing gases which obey gas equation approximately only under two conditions

- (i) Very low pressure
 (ii) High temperature

To test that whether the relation $PV = nRT$ is valid for real gases, graphical representation of Boyle's Law was examined for various gases like CO, CH_4 , He and H_2 etc.

A plot of $PV - P$ at constant temperature for a number of gases show deviation from ideal behaviour. Therefore, $PV = nRT$ cannot be applied to these gases. Thus, another equation must be sought in order to correlate P , V , T for these gases; which is van der Waals equation.

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT, \text{ where } a \text{ and } b \text{ are van der Waals constants}$$



Causes of Deviation

There are two hypothetical postulates in the kinetic theory of gases. These are as follows

- (1) The volume of a molecule is negligible as compared to total volume of the gas. Actually, gas molecules do possess some volume which account for the deviation.
 (2) There is no intermolecular forces of attraction between gaseous molecules.

(Either forces of attraction or repulsion exists between gaseous molecules, which may be small in magnitude).

By correcting these two postulates, we get an equation which can be applied to the gases which deviate from ideal behaviour. This deviation of a gas from ideal behaviour can also be expressed in terms of compressibility factor (Z).

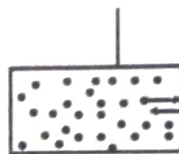
$$Z = \frac{PV}{RT} \text{ [for 1 mole]} \quad \text{or} \quad Z = \frac{(V_m)_{\text{real}}}{(V_m)_{\text{ideal}}}, \text{ where}$$

V_m is the molar volume

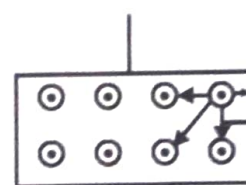
For ideal gas, $Z = 1$

For real gas, $Z > 1$ or $Z < 1$

Pressure Correction (correction due to intermolecular forces)



Ideal



Real

where P = observed pressure of the gas.

P' = pressure correction (pressure loss due to intermolecular forces of attraction and decrease in number of collisions on wall)

$$P' \propto \frac{n^2}{V^2}$$

$$P' = a \left(\frac{n}{V} \right)^2 \text{ (where } a \text{ is van der Waals constant)}$$

$$P' = a \frac{n^2}{V^2} \text{ (unit of } a = \text{atm litre}^2 \text{ mol}^{-2}\text{)}$$

$$\therefore P_1 = \left(P + a \frac{n^2}{V^2} \right) = \left(P + \frac{a}{V_m^2} \right) \quad \dots(i)$$

Volume Correction (Correction due to volume of gas molecules)

Ideal gas equation assumes molecules to be point masses. But at high pressure we have learnt that the volume of the gas molecules cannot be neglected in comparison to the volume of the container in which gas is kept. So, volume correction is applied. 'V' term in the ideal gas equation is the volume of the vessel in which gas molecules are free to move. But as the molecules themselves take up space, so now molecules instead of moving in volume 'V' are only restricted to volume (V - nb). nb is the volume which is incompressible as molecules are impenetrable spheres.

$$V_{\text{corrected}} = V - nb \quad \dots(ii)$$

where, b is van der Waals constant (unit = L mol⁻¹)
n is number of moles of gas

van der Waals Equation

$$\text{Ideal gas equation } P_{\text{ideal}} V_{\text{ideal}} = nRT \quad \dots(iii)$$

Putting the value of P_{ideal} and V_{ideal} from equation (i) and equation (ii) in equation (iii)

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT \text{ (van der Waals equation for real gases)}$$

Magnitude of 'b'

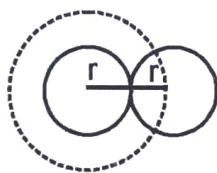
$$\text{Volume of dotted sphere} = \frac{4}{3} \pi (2r)^3 = 8 \times \frac{4}{3} \pi r^3$$

$$\text{If one molecule is taken, then it is } 4 \times \frac{4}{3} \pi r^3$$

If N_A molecules are taken, then

$$b = 4 \times \frac{4}{3} \pi r^3 N_A$$

$$b = 4V_m$$

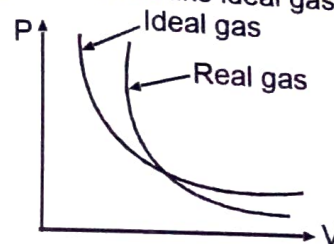


where V_m is the volume of 1 mole of molecules (and not the volume of container).

Excluded volume is equal to four times the actual volume possessed by a single molecule.

Knowledge Cloud

- (i) The graph indicates that at very low pressure real gas behaves like ideal gas.



(ii) Gas	Volume at STP
Ar	22.37
CO ₂	22.54
N ₂	22.69
O ₂	22.69
H ₂	22.72
Ideal	22.71

COMPRESSIBILITY FACTOR

Compressibility factor is the factor which decides the extent of deviation of real gases from the ideal gas behaviour.

Compressibility factor 'Z' is defined as the ratio of PV to nRT, mathematically

$$Z = \frac{PV}{nRT}$$

A graph is plotted for different gases between compressibility factor and over a range of pressure.

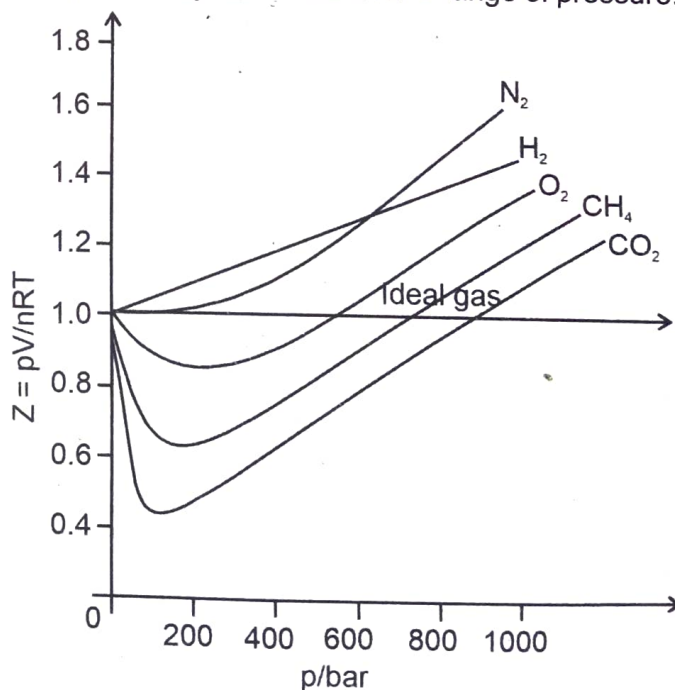


Fig.: Variation of compressibility factor for some gases

Conclusions made from the graph :

- (i) **$Z = 1$** : Compressibility factor is equal to unity for ideal gases at all temperature and pressure values.

Reason : For ideal gases, gas equation is $PV = nRT$. As PV is equal to nRT , so their ratio is unity.

- (ii) **$Z \approx 1$** : Compressibility factor is almost equal to unity at **very low pressure**.

Reason : At very low pressure, the interactions between the molecules is negligible, therefore there is no appreciable effect on the volume. So, $Z = 1$

- (iii) **$Z > 1$** : Compressibility factor is greater than unity (positive deviation) at **high pressure**.

Reason : At high pressure, the gas molecules come very close to each other, so the forces of repulsion start operating between them. For ideal gases, as the pressure increases,

volume decreases proportionally ($P \propto \frac{1}{V}$) so the product PV remains constant. But this is not the case with real gases. In them, with increase in pressure (high pressure) the forces of repulsion does not allow the volume to decrease proportionally. So, the product PV also increase with increasing pressure ($PV > 1$).

It can be better understood from the following derivation.

$$Z = \frac{PV_{\text{real}}}{nRT} \quad \dots(i)$$

Gas showing ideal behaviour

$$\left(V_{\text{ideal}} = \frac{nRT}{P} \right) \quad \dots(ii)$$

Putting the value of $\frac{nRT}{P}$ from equation (ii) in equation (i), we get

$$Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}$$

So, the Z value depends upon the ratio of actual volume of the gas to the volume of ideal gas at that temperature.

- (iv) **$Z < 1$** : Compressibility factor is less than unity (Negative deviation) when **pressure is intermediate**.

Reason : When the pressure is intermediate, then the gas molecules are at a sufficient distance to avoid repulsion. Instead, forces of attraction operate. Because of this, the molecules attract each other and they come

closer. So, the volume decreases more than expected or decrease in volume is not proportional to the increase in pressure. Thereby the product PV decreases with increasing pressure. ($PV < 1$). So, the ratio of

$$\frac{PV}{nRT} \text{ becomes less than 1.}$$

At such pressure condition, the gases are **more compressible** due to forces of attraction.

So, it is concluded **that behaviour of gases becomes ideal when pressure is very low**. Thus gases show ideal behaviour when the volume occupied is large so that the volume of the molecules can be neglected in comparison to it.

	Compressibility Factor Z	Pressure	Compressibility
(1)	$Z = 1$	All	Normal
(2)	$Z \approx 1$	Low	Normal
(3)	$Z > 1$	High	Difficult
(4)	$Z < 1$	Intermediate	Easy

Pressure alone does not decide the behaviour of gases. **Temperature is also an important factor which plays role.**

Boyle Temperature or Boyle Point : It is the temperature at which real gases obey ideal gas laws over an appreciable range of pressure.

Boyle temperature depends on the nature of the gas.

Effect of Temperature on the Compressibility Factor :

- (i) **Above the Boyle temperature** : Real gas show positive deviation ($Z > 1$) from ideality. This is because with increase in temperature, the molecules move far from each other. So, volume increases thereby the forces of attraction between the molecules become feeble.

- (ii) **Below the Boyle temperature** : Below the Boyle temperature ' Z ' value first decreases and reaches a minimum value with increase in pressure because of forces of attraction which start operating between the molecules. Later, on further increase in pressure force of repulsion operate. So, now value of Z increases continuously.

Conclusions

Gases show ideal behaviour at

- High temperature
- Very low pressure

Illustration 6

The volume of gas A is twice than that of gas B. The compressibility factor of gas A is thrice than that of gas B at same temperature. The pressure of the gases for equal number of moles are

- (1) $P_A = 2P_B$ (2) $P_A = 3P_B$
 (3) $3P_A = 2P_B$ (4) $2P_A = 3P_B$

Solution : $Z = \frac{PV_m}{RT}$

$$\therefore \frac{Z_A}{Z_B} = \frac{P_A V_A}{P_B V_B}$$

$$3 = \frac{P_A}{P_B} \times 2$$

$$2P_A = 3P_B$$

Answer (4)**LIQUEFACTION OF GASES**

Gases can be liquefied by applying high pressure and by cooling.

Critical Parameters

Critical Temperature : It is the temperature above which gas cannot be liquefied, no matter how high be pressure.

$$T_c = \frac{8a}{27Rb}$$

Critical Pressure : It is the minimum pressure that is required to liquefy a gas at critical temperature.

$$P_c = \frac{a}{27b^2}$$

Critical Volume : It is the volume occupied by gas at critical temperature and critical pressure.

$$V_c = 3b$$

CRITICAL PARAMETERS OF A REAL GAS

$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$ can be expanded as

$$PV_m - Pb + \frac{a}{V_m} - \frac{ab}{V_m^2} - RT = 0$$

$$\text{or } V_m^3 - \left(\frac{RT}{P} + b\right)V_m^2 + \frac{a}{P}V_m - \frac{ab}{P} = 0 \quad \dots(i)$$

(Cubic equation in volume)

This equation has three roots.

At critical condition, all three roots are equal.

$$\text{So, } V_1 = V_2 = V_3 = V_c = V_m$$

$$\text{or } (V_m - V_c)^3 = 0$$

$$\text{or } V_m^3 + 3V_m V_c^2 - 3V_m^2 V_c - V_c^3 = 0 \quad \dots(ii)$$

Comparing coefficients of (i) and (ii), we get

$$V_c^3 = \frac{ab}{P_c}, \quad \frac{RT_c}{P_c} + b = 3V_c, \quad \frac{a}{P_c} = 3V_c^2$$

Solving, we get

$$V_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

Critical compression factor, $Z_c = \frac{P_c V_c}{RT_c} = \frac{3}{8}$ for van der Waals gas.

Andrews Isotherm Curve

It is the graph plotted between pressure and volume during liquifaction of a gas.

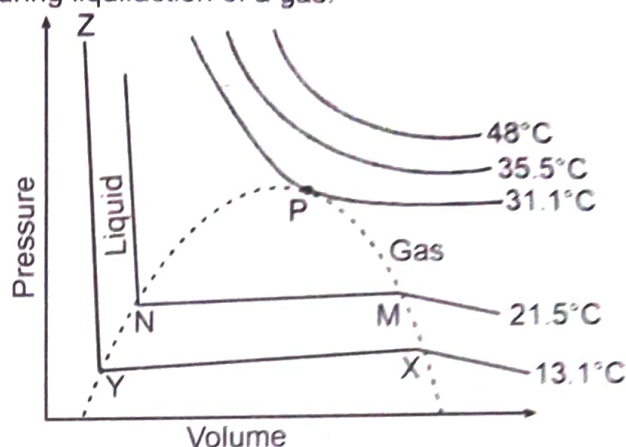


Fig. : Isotherm of CO_2 showing critical region

At the critical point P, the three roots of van der Waals equation are not only real and positive but also identical and equal to the critical volume V_c .

Inversion Temperature (T_i) : If a compressed gas is allowed to pass through a fine hole, it may show cooling or heating or no change in temperature depending upon the temperature of gas. For each gas, there exists a characteristic temperature known as inversion temperature (T_i).

If $T > T_i$ heating is produced

If $T < T_i$ cooling is produced

If $T = T_i$ neither heating nor cooling

$$T_i = \frac{2a}{Rb} = 2T_b$$

Illustration 7

Given below are the critical temperatures of some of the gases :

Gas	Critical temperature (K)
He	5.2
CH_4	190.0
CO_2	304.2
NH_3	405.5

The gas showing least adsorption on a definite amount of charcoal is

- (1) He (2) CH_4
 (3) CO_2 (4) NH_3

Solution : Extent of adsorption $\propto T_c$ (critical temperature)

$\therefore$ Lower the T_c , Lower will be the adsorption

Hence, Helium shows least adsorption on a definite amount of charcoal. **Answer (1)**

EXERCISE

22. Compressibility factor for an ideal gas is
 (1) 1.5 (2) 1.0
 (3) 2.0 (4) ∞
23. A real gas will approach ideal behaviour at
 (1) Low temperature and low pressure
 (2) High pressure and low temperature
 (3) High temperature and high pressure
 (4) High temperature and low pressure
24. At low pressure, the van der Waals equation is written as $(P + \frac{a}{V_m^2})V_m = RT$.
 The compressibility factor is then equal to
 (1) $(1 - \frac{a}{RTV_m})$ (2) $(1 - \frac{RTV_m}{a})$
 (3) $(1 + \frac{a}{RTV_m})$ (4) $(1 + \frac{RTV_m}{a})$
25. Boyle temperature is given by
 (1) $T_B = \frac{a}{Rb^2}$ (2) $T_B = \frac{a}{Rb}$
 (3) $T_B = \frac{a}{27b^2}$ (4) $T_B = \frac{b}{aR}$
26. The critical constants for water are 374°C , 218 atm and $0.0566 \text{ L mol}^{-1}$. Then
 (1) van der Waals constant $a = 0.189 \text{ L mol}^{-1}$
 (2) van der Waals constant $b = 0.0189 \text{ L mol}^{-1}$
 (3) van der Waals constant $b = 2.095 \text{ L mol}^{-1}$
 (4) At critical point $z = \frac{8}{3}$

GAS ANALYSIS OR EUDIOMETRY

Eudiometry is based on Avogadro's law which states that equal volume of all gases under similar conditions of temperature and pressure contain equal number of molecules (or equal number of moles).

During gas analysis, the Eudiometer tube filled with mercury is inverted over a trough containing mercury. A known volume of the gas or gaseous mixture is introduced, which displaces an equivalent amount of Hg. Now excess of oxygen is introduced and is electrically sparked, whereby the combustible material gets oxidised. The volume of CO_2 , H_2O or other gaseous products of combustion are next determined by absorbing them in suitable reagent.

Assumptions used during Eudiometric calculations :

- If temperature is not given, then assume it as room temperature (25°C)
- Volume mentioned is for gases.

- N_2 gas by definition is inert.
- Whenever volume of gases are mentioned volume of H_2O vapour is ignored.
- We must know the solution which absorbs gases.
 - Acidic gases CO_2 , SO_2 , NO_2 , X_2 by NaOH or KOH solution
 - CO , C_2H_2 etc. $\rightarrow$ Ammonical Cu_2Cl_2
 - O_3 $\rightarrow$ Turpentine oil
 - NO $\rightarrow$ FeSO_4 solution
 - N_2 $\rightarrow$ Heated Mg
 - O_2 $\rightarrow$ Alkaline pyrogallol
 - NH_3 by acids and CuSO_4 solution.

Water vapour produced during the reaction can be determined by noting contraction in volume caused due to cooling.

From the data, number of useful conclusions regarding reactions amongst gases can be drawn.

- Volume - volume relationship amongst gases or simple gaseous reactions.
- Composition of gaseous mixture.
- Molecular formulae of gases.

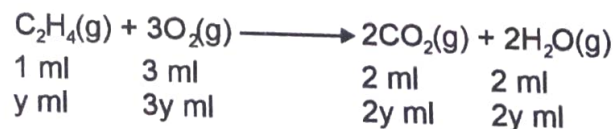
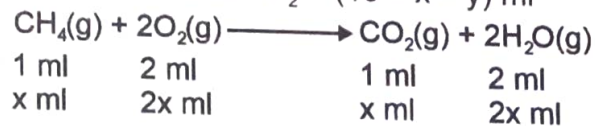
Example 8 : 10 ml of a mixture of CH_4 , C_2H_4 and CO_2 was exploded with excess of air. After explosion there was a contraction in volume of 17 ml and after treatment with KOH there was a further reduction of 14 ml. What was composition of the mixture? (Volume measurement refer to the same P - T conditions)

Solution : In the explosion, reactant CO_2 does not change, while CH_4 and C_2H_4 changes to CO_2 and H_2O .

Let the volume of $\text{CH}_4 = x \text{ ml}$

Volume of $\text{C}_2\text{H}_4 = y \text{ ml}$

$\therefore$ volume of $\text{CO}_2 = (10 - x - y) \text{ ml}$



After the explosion there was contraction of 17 ml. It means, when the explosion took place at that time water is in gaseous form, after the explosion, water vapours are converted into liquid form and the volume of liquid water is assumed to be zero (in comparison to gases).

$$\therefore 2x + 2y = 17$$

...(i)

Since KOH absorbs CO_2 gas when the gaseous mixture is treated with KOH, further reduction in volume is 14 ml, therefore volume of CO_2 in gaseous mixture is 14 ml.

$$10 - x - y + x + 2y = 14 \quad \dots(ii)$$

$$10 + y = 14$$

or $y = 4 \text{ ml}$

On putting $y = 4$ in equation (i), we get

$$x = 4.5$$

$\therefore$ Volume of $\text{CH}_4 = 4.5 \text{ ml}$

and Volume of $\text{C}_2\text{H}_4 = 4.0 \text{ ml}$

and Volume of $\text{CO}_2 = (10 - 4 - 4.5) = 1.5 \text{ ml}$

LIQUID STATE

It is the intermediate state between gaseous and solid states.

Liquids possess fluidity like gases but incompressibility like solids.

Properties of liquid are :

- (i) A liquid is made up of molecules. Only Hg(l) is in atomic state.
- (ii) The intermolecular forces of attraction in a liquid are quite large.
- (iii) Liquids have no definite shape but have definite volume as the cohesive forces are strong.
- (iv) Liquids diffuses slowly in comparison to gas.
- (v) **Evaporation** : The process of change of liquid into vapour state at any given temperature is evaporation.

Evaporation is accompanied by cooling as average kinetic energy of remaining molecules decreases.

The rate of evaporation depends upon the strength of intermolecular forces (cohesive forces).

The liquid having low intermolecular forces evaporate faster.

Example: Ether evaporates faster than alcohol.

Alcohol evaporates faster than water.

The order of intermolecular forces is Ether < Alcohol < Water

Rate of Evaporation $\propto$ Surface area.

Rate of Evaporation $\propto$ Temperature.

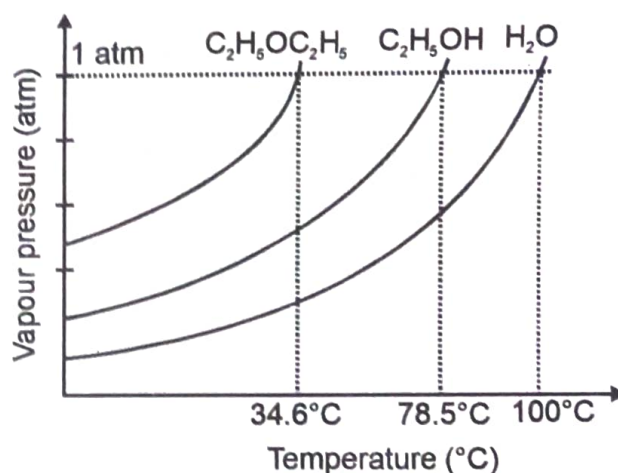
Flow of air helps the molecules to go away from the surface of liquid and therefore, increases the evaporation

(vi) **Vapour Pressure** : In a closed vessel when the rate of evaporation become equal to rate of condensation, i.e. equilibrium is established, the pressure exerted by the vapours of liquid on its surface is known as vapour pressure.

The magnitude of vapour pressure depends upon the following points :

- (a) **Nature of Liquid** : The vapour pressure of a liquid depends upon the nature of the liquid. Lesser are the intermolecular forces between the liquid molecules more is the equilibrium vapour pressure.
- (b) **Temperature of Liquid** : As the temperature increases, more number of molecules possess above-average kinetic energies, so the number of molecules escaping from the liquid surface increases. Hence, vapour pressure of a liquid increases with rise in temperature.

Note : Increase in surface area has no effect on vapour pressure.



Clausius Clapeyron Equation

$$\log \frac{P_2}{P_1} = \frac{\Delta H}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

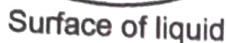
(vii) **Boiling Point** : Boiling point of the liquid is the temperature at which the vapour pressure of the liquid is equal to the atmospheric pressure.

Normal Boiling Point : As the atmospheric pressure varies with altitude and other conditions, the boiling points are reported at 1 atm. So, the normal boiling point of a liquid is the temperature at which the vapour pressure of the liquid is 1 atm.

e.g. Normal boiling point of water = 100°C

e.g. Standard boiling point of water = 99.6°C
Standard boiling point of the liquid is slightly lower than the normal boiling point of the liquid.

The property of surface tension may also be described in terms of the tendency of a liquid to decrease its surface area.



where F is the common magnitude of the forces exerted on each other by the two parts of the surface across a line of length l .

The forces between the layers which oppose the relative motion between them are known as the forces of viscosity.

A diagram of a cylindrical control volume in a fluid flow. The cylinder has a cross-sectional area A and length dz . The fluid velocity is $v + dv$. The vertical distance from the bottom to the center is z .

The $-ve$ sign indicates that force is frictional and opposes relative motion.

$$\phi = \frac{1}{\eta}$$

EXERCISE

27. When a drop of liquid splits up into a number of drops
 - (1) The volume increases
 - (2) Energy is liberated
 - (3) The surface area increases and energy is absorbed
 - (4) The area decreases and no energy exchanged
28. What would be percentage composition by volume of a mixture of CO and CH₄, whose 10.5 mL requires 9 mL oxygen for complete combustion?
 - (1) CO = 80% CH₄ = 20%
 - (2) CO = 90% CH₄ = 10%
 - (3) CO = 76.2% CH₄ = 23.8%
 - (4) CO = 66% CH₄ = 34%
29. 20 mL mixture of CO and C₂H₄ was exploded with 50 mL of O₂. The volume after explosion was 45 mL. On shaking with NaOH solution, only 15 mL O₂ was left behind. What were the volume of CO and C₂H₄ in mixture?
 - (1) CO = 5 mL C₂H₄ = 15 mL
 - (2) CO = 10 mL C₂H₄ = 10 mL
 - (3) CO = 8 mL C₂H₄ = 12 mL
 - (4) CO = 15 mL C₂H₄ = 5 mL
30. 12 mL of a gaseous hydrocarbon was exploded with 50 mL of oxygen. The volume after explosion was 32 mL. On treatment with KOH the volume reduce to 8 mL. Then formula of hydrocarbon is
 - (1) C₂H₄ (2) C₂H₂
 - (3) C₃H₈ (4) C₂H₆
31. Which of the following property of liquid always increase on increasing temperature?
 - (1) Vapour pressure (2) Surface tension
 - (3) Viscosity (4) Both (1) & (2)

Additional Information

1. Comparison of van der Waals gas equation and virial equation.
The van der Waals equation for 1 mole gas is

$$\left(P + \frac{a}{\bar{V}^2}\right)(\bar{V} - b) = RT$$

where $\bar{V}$ is the molar volume

$$\text{or } \left(P + \frac{a}{\bar{V}^2}\right)\bar{V}\left(1 - \frac{b}{\bar{V}}\right) = RT$$

$$\left(P\bar{V} + \frac{a}{\bar{V}}\right) = \frac{RT}{\left(1 - \frac{b}{\bar{V}}\right)} = RT\left(1 - \frac{b}{\bar{V}}\right)^{-1}$$

Since, $\frac{b}{\bar{V}} \ll 1$, so we can expand the first term using $(1 - X)^{-1} = 1 + X + X^2 + \dots$

$$\left(P\bar{V} + \frac{a}{\bar{V}}\right) = RT\left[1 + \frac{b}{\bar{V}} + \left(\frac{b}{\bar{V}}\right)^2 + \dots\right]$$

$$P\bar{V} = RT\left[1 + \frac{b}{\bar{V}} + \left(\frac{b}{\bar{V}}\right)^2 + \dots\right] - \frac{a}{\bar{V}}$$

$$\Rightarrow P\bar{V} = RT\left[1 + \left(\frac{b}{\bar{V}} - \frac{a}{RT\bar{V}}\right) + \left(\frac{b}{\bar{V}}\right)^2 + \dots\right]$$

$$= RT\left[1 + \frac{\left(b - \frac{a}{RT}\right)}{\bar{V}} + \left(\frac{b}{\bar{V}}\right)^2 + \dots\right] \quad \dots(i)$$

$$P\bar{V} = RT\left[1 + \frac{B}{\bar{V}} + \dots\right] \quad \dots(ii)$$

Comparing equation (i) with equation (ii), we get

$$B = \left(b - \frac{a}{RT}\right)$$

2.

This graph is expected theoretically which show cubic form of van der Waal's gas

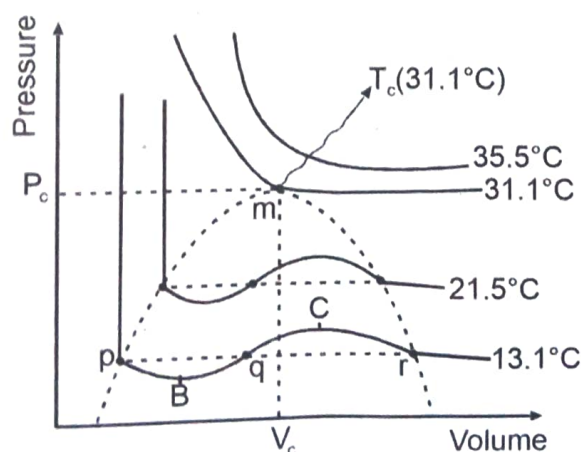


Fig. : Isotherm of CO_2 according to van der Waals equation